

COMPOSITE MATERIAL DATASHEETS CFRP GURIT 75-RC200T (PRELIMINAR DVs)

MATERIAL IDENTIFICATION

EU 1E001 - This data is controlled under EU Dual Use Regulation 2021/821, as amended. Authorization may be required for the export of the data outside of the EU.

Short informal name	RC200T/SE75-TW
Commercial name	Gurit SE75-T1/RC200T/1270/42+/-3%/1PO
Specifications	-
Description	Carbon Fiber fabric type Gurit 75-RC200T High Performance, hot melt, preimpregnated in epoxy resin for large open mould vacuum processes.
Version	0.7 Remarks: - This datasheet contains preliminary Design Values. The tests program for the mechanical characterization of this material has not concluded. - The applicability of these Design Values is restricted to use in the <i>SiRTAP</i> program (with very particular certification/qualification specifications)

REFERENCES

- [1] FIDAMC Screening Tests Data (MS Excel file "0 - Tejido - Mechanical resultsv11.xlsx")
[2] Airbus DS software COMALL v.0.24 files "Gurit_75-RC200T_23C_v4.*"
[3] TEA-AT-NT-210004 Is. D "Stress Methods and Material Design Values applicable to the Structural Analysis of the *SiRTAP* airframe"

GENERAL NOTE ABOUT BASIS VALUES

B-basis values are not presented in this datasheet. The penalizations to material design values required to account for dispersion in material strengths shall be discussed and agreed with certification/qualification authorities, and applied to structural analyses, either in the classical form of reduction of the mean-values, or by applying extra safety factors to applied loads (see [3] §1.4.2).

PHYSICAL PROPERTIES

Cured Ply Thickness (CPT)	0.253 mm	Coefficients of Thermal Expansion CTE (1/°C)	Range	-55...20°C	20...70°C	70...120°C
Density	- kg/m ³		α_1	-	-	-
Tg-onset (wet)	- °C		α_2	-	-	-
Max. Operational Limit (MOL)	70 °C		α_3	-	-	-

ELASTIC MODULI

Property / Units	E_1 (MPa)	E_2 (MPa)	G_{12} (MPa)	ν_{12}	K_B
Values	50500	50500	3300	0.05	-

Property / Units	E_3 (MPa)	G_{13} (MPa)	G_{23} (MPa)	ν_{13}	ν_{23}
Values	10400	2900	2900	0.5	0.5

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PLAIN STRENGTHS

Ultimate stresses of the ply (from UD-laminates testing).

Property	Mean Values (MPa)			B-basis K_B	E-KDF		B-basis Values (MPa)		
	RT/Dry	50°C/W	70°C/W		50°C/W	70°C/W	RT/Dry	50°C/W	70°C/W
F_{1t}	599	599	599	-	1.00 (*)	1.00 (*)	-	-	-
F_{1c}	602	547	505	-	0.908	0.838	-	-	-
F_{2t}	599	599	599	-	1.00 (*)	1.00 (*)	-	-	-
F_{2c}	602	547	505	-	0.908 (**)	0.838 (**)	-	-	-
F_{12}	100	84	75	-	0.846	0.752	-	-	-

Design values for checking the laminate strength using the modified Maximum Strain Criterion.

Property	Mean Values ($\mu\epsilon$)			B-basis K_B	E-KDF		B-basis Values ($\mu\epsilon$)		
	RT/Dry	50°C/W	70°C/W		50°C/W	70°C/W	RT/Dry	50°C/W	70°C/W
ϵ_{tall}	11900	11900	11900	-	1.00 (*)	1.00 (*)	-	-	-
ϵ_{call}	11900	10800	9900	-	0.908	0.838	-	-	-

(*) Assumed tension strengths of the ply in fibre directions are not significantly affected by ambient conditions

(**) Degradation due to ambient conditions in F2c (compression-weft) assumed equal to that for F1c (compression-warp)

INTERLAMINAR PLY STRENGTHS

Property	Mean Values			B-basis K_B	E-KDF		B-basis Values		
	RT/Dry	50°C/W	70°C/W		50°C/W	70°C/W	RT/Dry	50°C/W	70°C/W
F_{13} (MPa)	60	46	41	-	0.769	0.691	-	-	-
F_{3t} (MPa)	-	-	-	-	-	-	-	-	-
G_{IC} (N/mm)	-	-	-	-	-	-	-	-	-
G_{IIC} (N/mm)	-	-	-	-	-	-	-	-	-

OPEN HOLE AND BOLTED JOINTS

Reference material strengths in open-hole, filled-hole and bearing for the quasi-isotropic laminate.

Property	Mean Values (MPa)			B-basis K_B	E-KDF		B-basis Values (MPa)		
	RT/Dry	50°C/W	70°C/W		50°C/W	70°C/W	RT/Dry	50°C/W	70°C/W
OHT_{QI}	-	-	-	-	-	-	-	-	-
OHC_{QI}	-	-	-	-	-	-	-	-	-
FHT_{QI}	220	220	220	-	1.000	1.000	-	-	-
FHC_{QI}	389	351	310	-	0.903	0.798	-	-	-
F_{BR-QI}	462	355	344	-	0.768	0.745	-	-	-